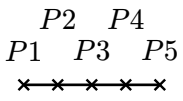
Label collision avoidance

Part 1 — numeric assertions

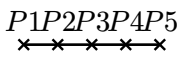
All numeric assertions passed.

Part 2 — visual cases

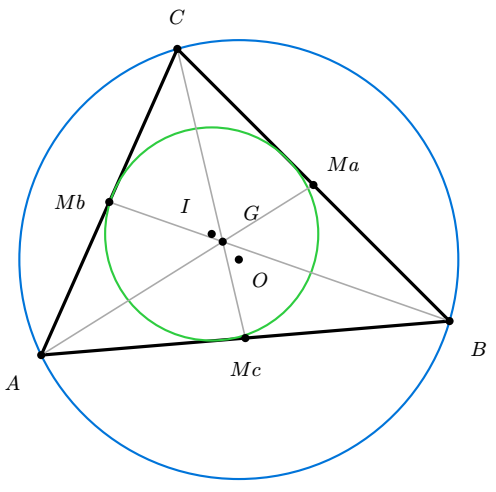
Five crowded points all requesting "above" (labels dodge the line, points and each other)



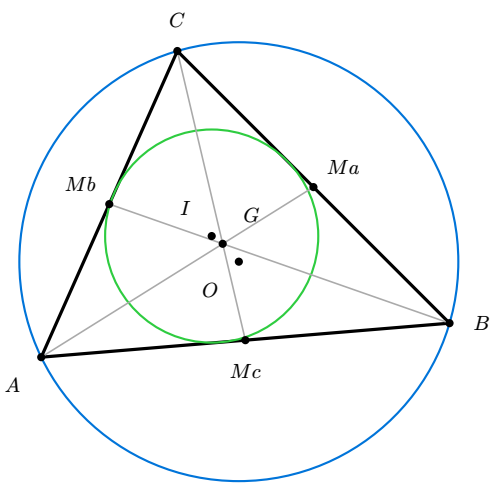
Same figure with avoidance disabled via ctz-label-avoid(false) (labels overlap — for comparison)



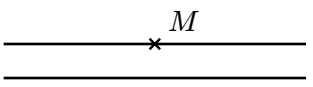
Crowded triangle: circumcircle + incircle + medians, everything labeled



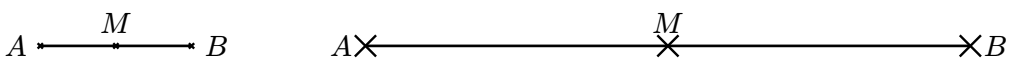
Same triangle, fully automatic (labels: true — no positions specified)



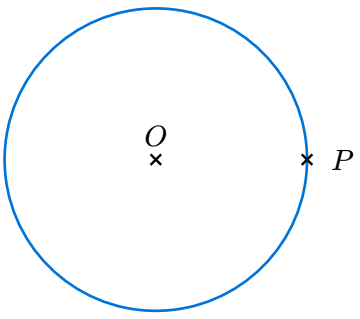
Point on a line, label "below" also blocked by a second line → shifted/shrunk



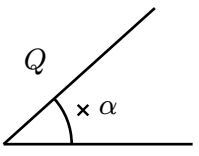
Scale independence: same figure at length 0.5cm and 2cm



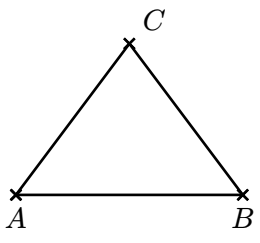
Labels avoid a circle stroke but are allowed inside the circle



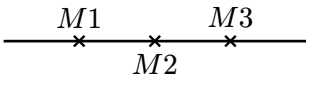
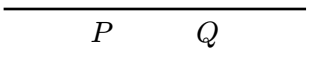
Angle label avoidance: α slides past the Q marker; Q's label dodges both



Duplicate labels are skipped (path auto-labels + explicit ctz-draw-labels)



Per-label granularity: M2 pinned exactly where asked (avoid: false), M1/M3 dodge



Turning avoidance back on mid-canvas